

Drawing Borders

Problem ID: borders

TODO: Hier sollte man sich natürlich auch noch ne Story ausdenken.

You are given n points that are coloured red, blue or green. Your task is to draw two polygons such that any two points of different colour are separated. The polygons may not touch or intersect each other and no point may lie on their boundary.

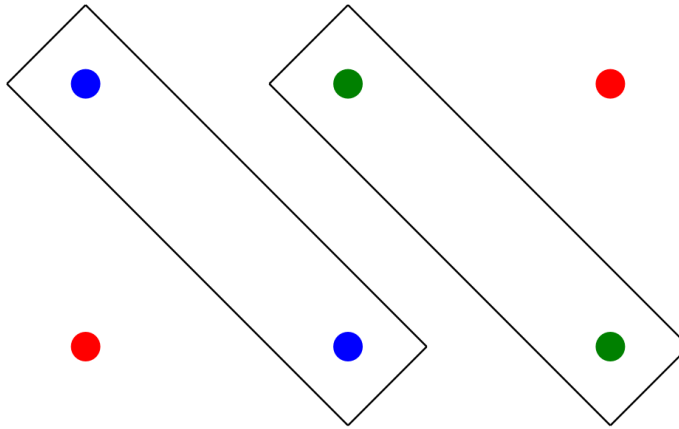


Figure 1: Illustration of the sample.

Input

The input consists of:

- One line with an integer n ($3 \leq n \leq 100$), the number of points.
- n lines, each with three integers x, y, c ($-1\,000 \leq x, y \leq 1\,000, 1 \leq c \leq 3$), representing a point at point at coordinates (x, y) of colour c ($1 = \text{blue}, 2 = \text{red}, 3 = \text{green}$). All positions are unique. There is at least one point of each colour.

Output

If there is no solution, print `impossible`. Otherwise, print the two polygons in the following format:

- One line with an integer m ($3 \leq m \leq 1\,000$), the number of vertices of the polygon.
- m lines, each with two real numbers x, y ($-2\,000 \leq x, y \leq 2\,000$), the vertices of the polygon in either clockwise or counter-clockwise order. The numbers may be given with up to three decimal places.

Sample Input 1

```
6
0 0 2
0 1 1
1 0 1
1 1 3
2 0 3
2 1 2
```

Sample Output 1

```
4
-0.3 1.0
1.0 -0.3
1.3 0.0
0.0 1.3
4
0.7 1.0
2.0 -0.3
2.3 0.0
1.0 1.3
```